

Chess - User documentation

How to install:

First download the gitlab project to your computer from this URL:

<https://gitlab.mff.cuni.cz/hofmanl3/chess-c-sharp>

After you download the project, open it.

In the Application folder run the Chess.exe file.

This should run your application, Enjoy!

How to play:

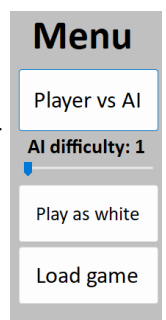
Upon starting the application you are greeted with a Main Menu.

Main Menu interface

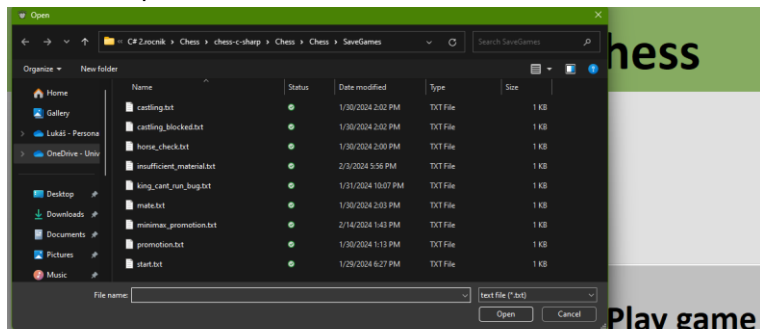


The Main Menu has several buttons:

- Player vs player:
 - This buttons decides who is going to be your opponent
 - If „Player vs player“ is visible, then you are going to play against another player
 - If „Player vs AI“ is visible, then you are going to play as only one color and the other is going to be played by the minimax algorithm.
 - Additionally, if Player vs AI is selected, new options are added:
 - A trackbar appears that changes the difficulty of the AI.
 - The lower the level is, the faster the decisions of the minimax will be, but also the better it will play.



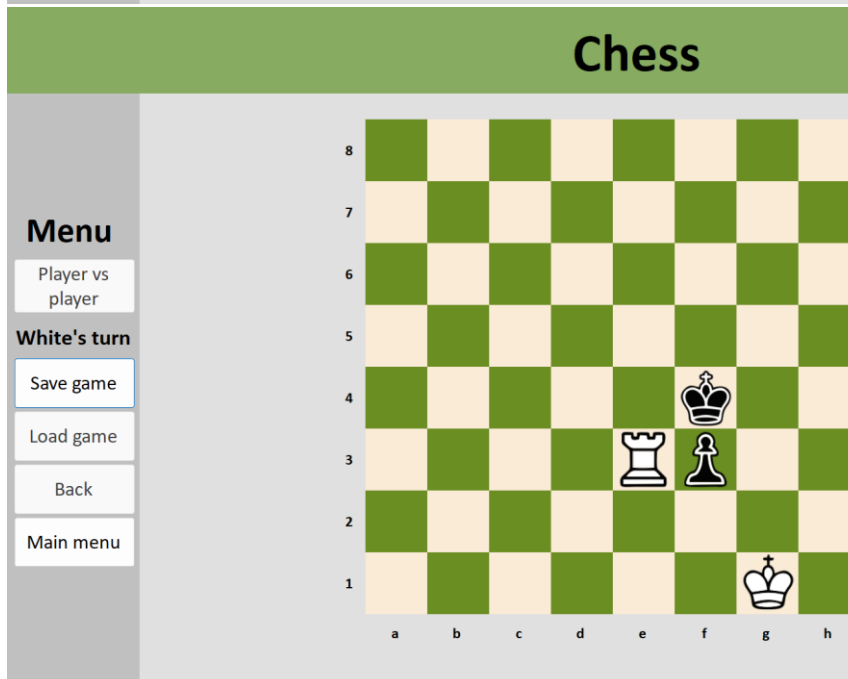
- A new button with the caption „Play as white“ appears – this button selects which color you want your pieces to be.
 - If Play as white is active, you are going to play as the white pieces
 - If Play as black is active, you are going to play as the black pieces
- Load game
 - This button opens a FileDialog where you can select a .txt save file of your previous games. Careful, it doesn't check if you selected the correct file, so be sure to select only saves from previous games!
 - When you select a save and open it, it will load and start the game for you. Here is an example:



Load game

Exit game

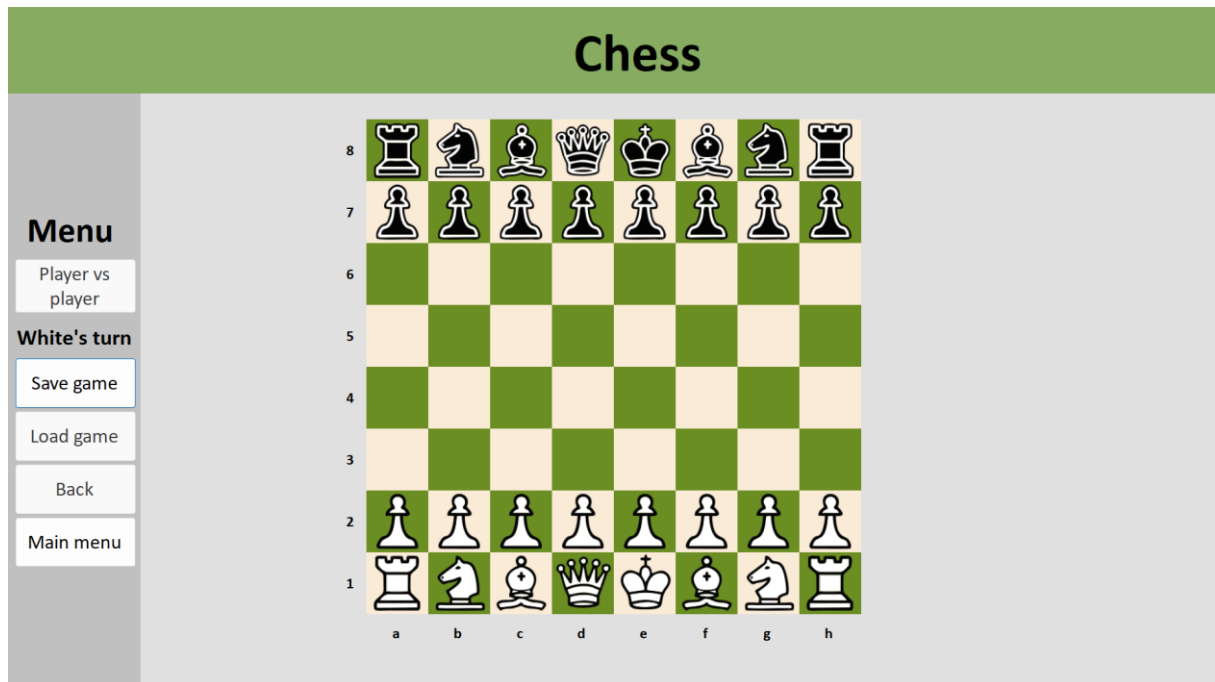
○



- Exit game
 - closes the application

- Play game
 - o starts a brand new game

Game interface



In the game interface except the game board a few new buttons appear.

- If you chose the Player vs player option
 - o Player vs player is now disabled as you can't change your opponent during game
 - o You can see whose turn it is
- Save game
 - o Works similar as the Load game button, but instead of loading saves the current game to a .txt file
- The Load button is disabled during the game, so you can't use it.
- Back
 - o It is initially disabled until you make a move. There is a history of your moves during that session of the game and you can use the Back button until you trace back all the way to the start of your loaded game.

Chess

Player vs
player

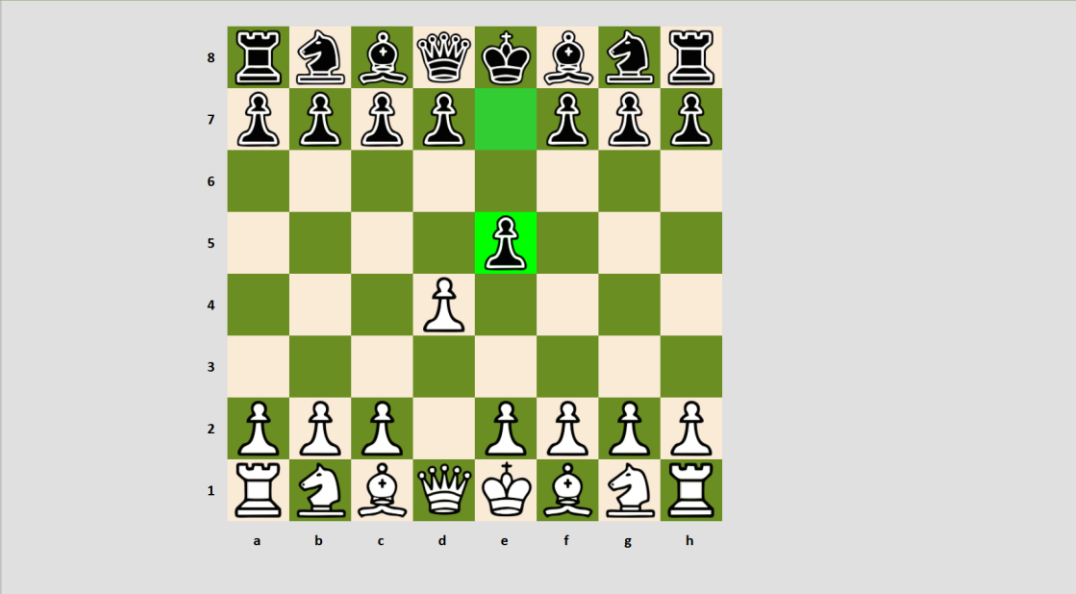
White's turn

Save game

Load game

[Back](#)

[Main menu](#)



Chess

Menu

Player vs player

Black's turn

Save game

Load game

Back

Main menu

Player vs
player

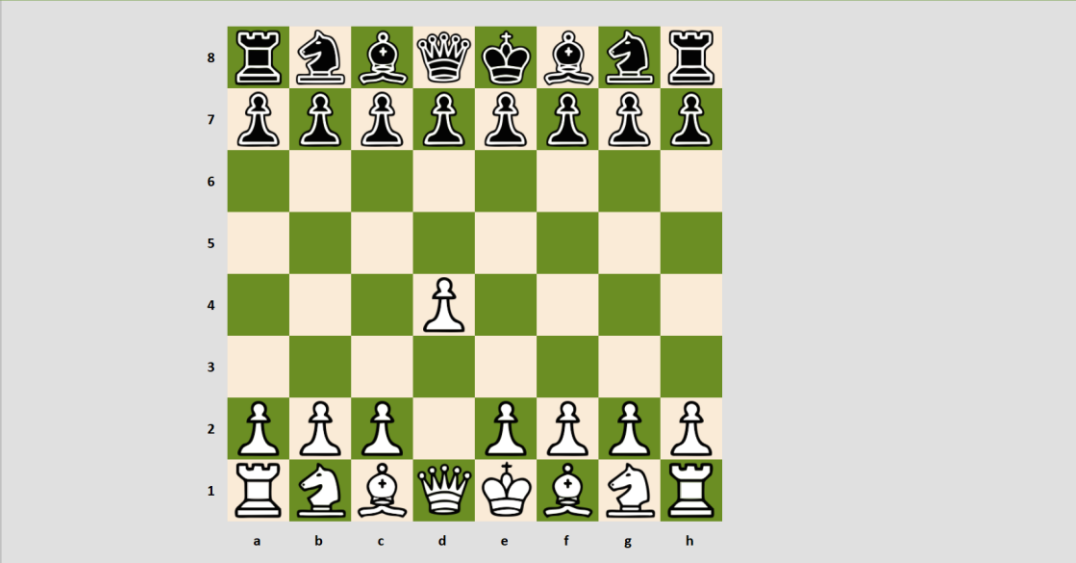
Black's turn

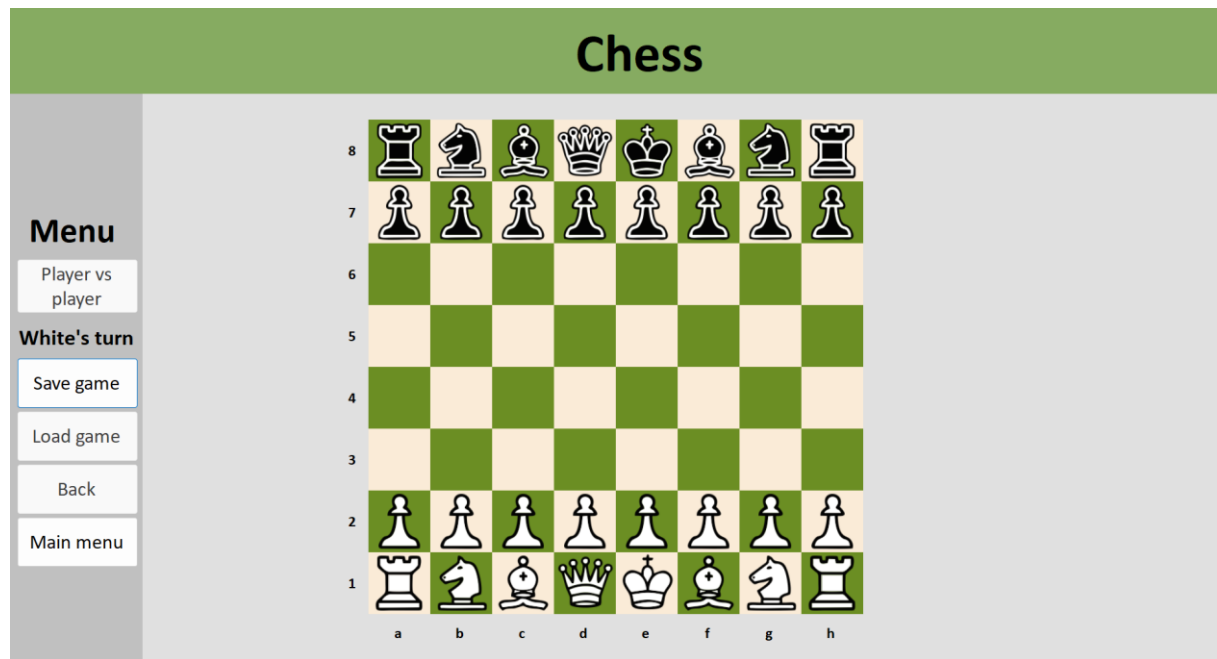
Save game

Load game

[Back](#)

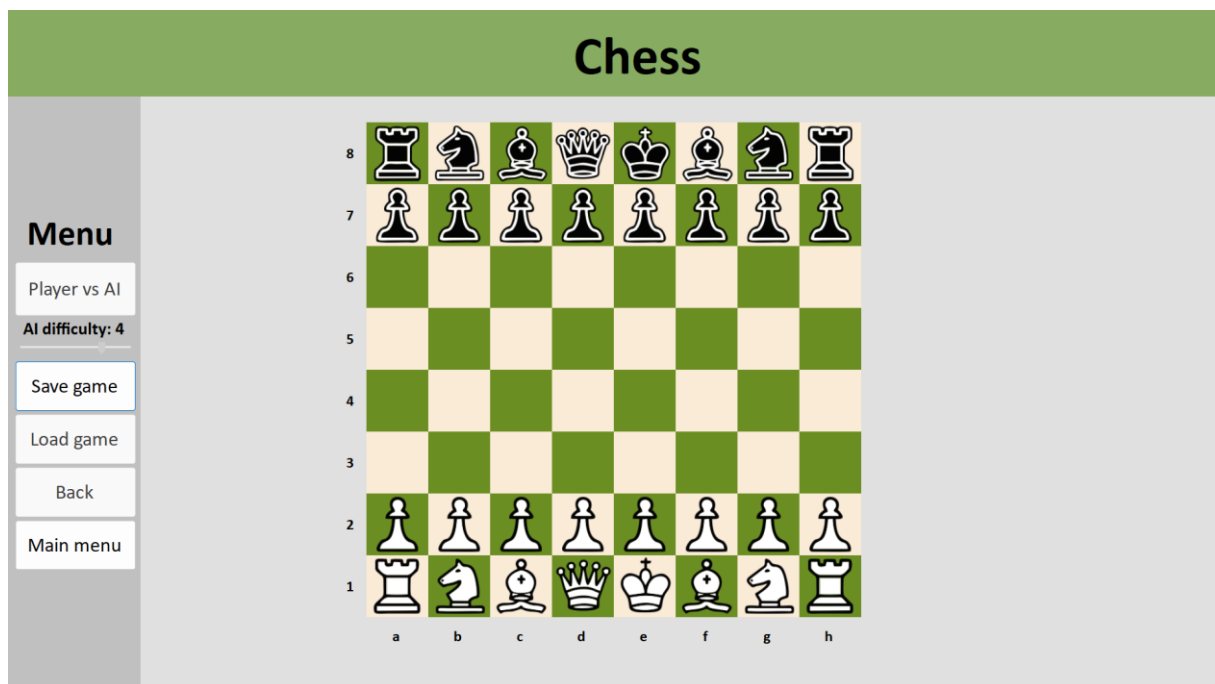
[Main menu](#)



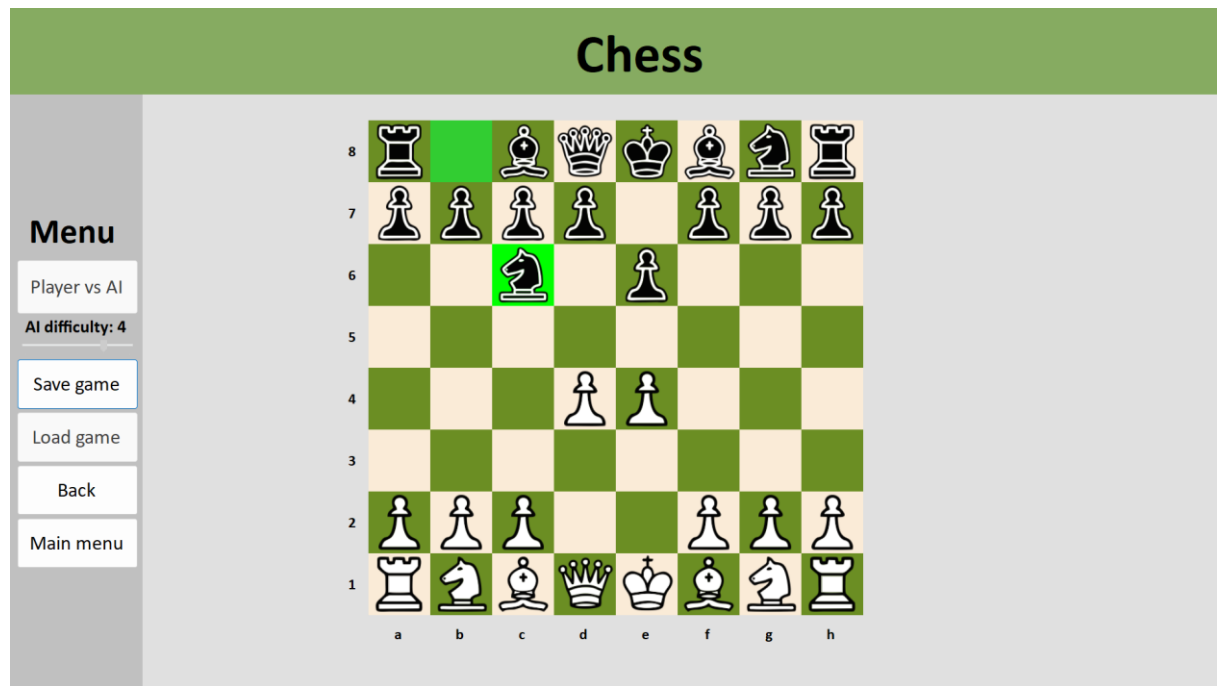


- Main menu
 - It switches the interface back to Main menu

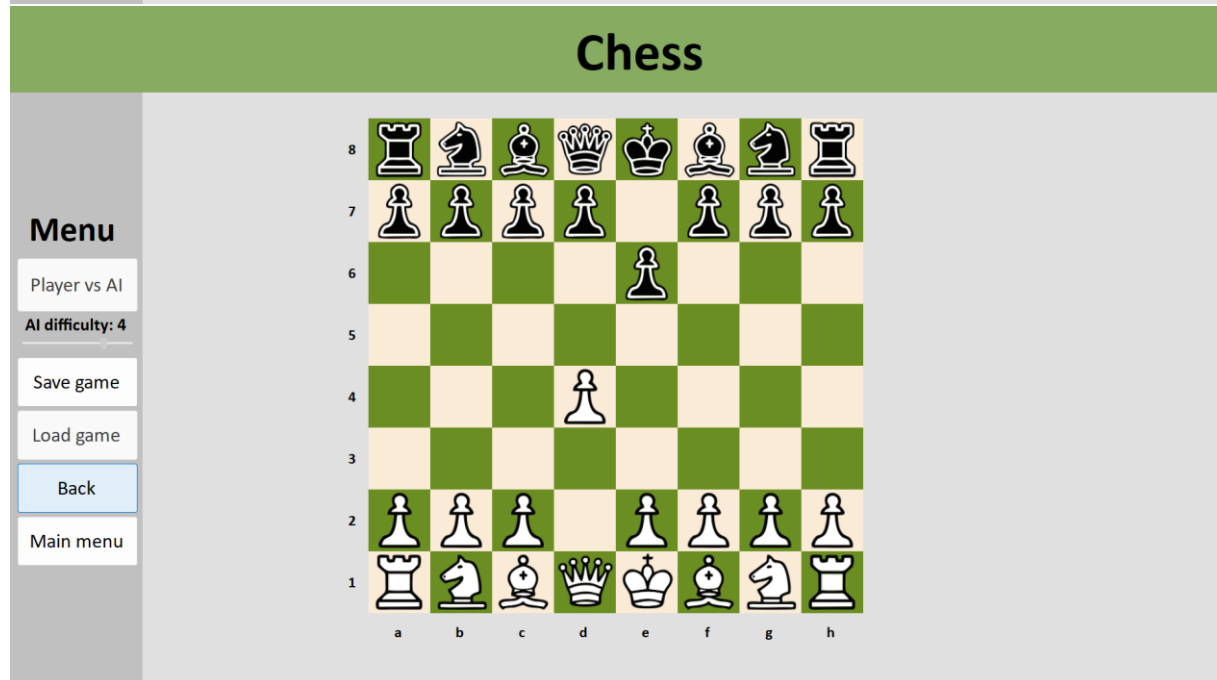
If you chose the Player vs AI option the interface is very similar:



- Again Player vs AI is disabled as you can't switch opponents during the game
 - You can't see whose turn it is as you chose your color before the start of the game and that won't change
- AI difficulty is shown during the game
 - You can't change the difficulty during the game, the trackbar is disabled
- The Back button works a tiny bit different – it rollbacks your last move, not the last move made.



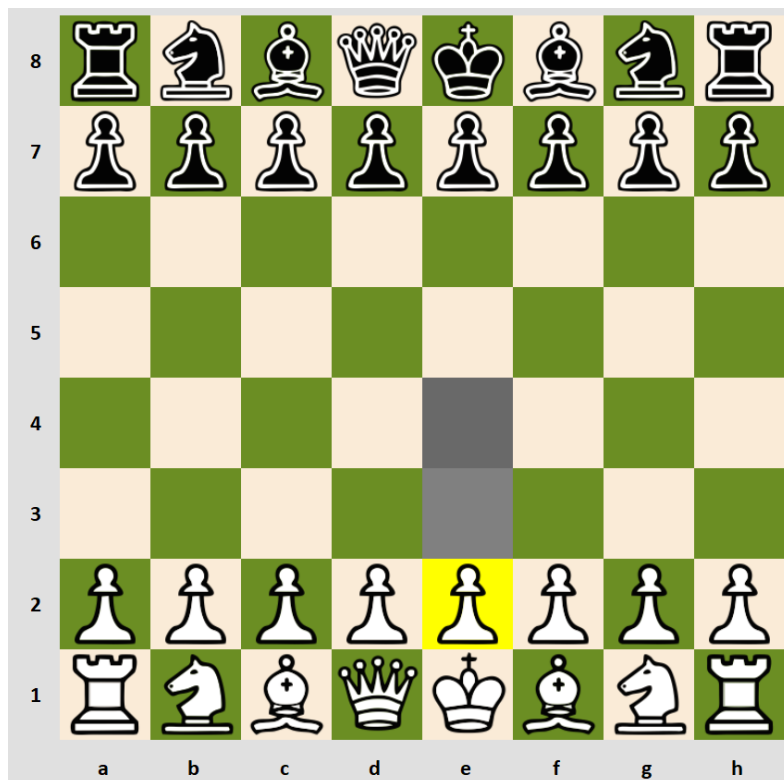
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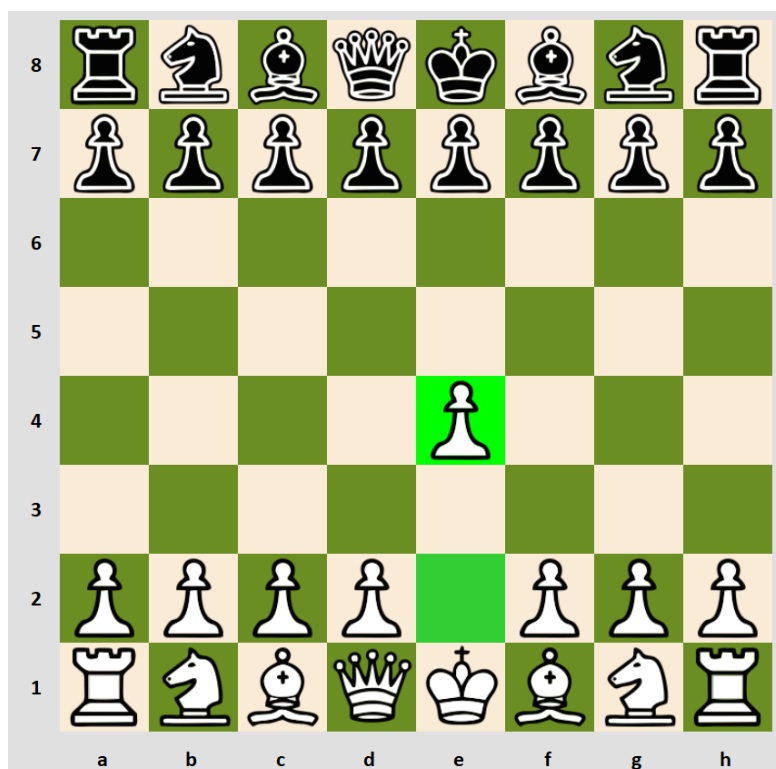
- The rest of the buttons remain the same as in the Player vs player version

How to make a move

Click on a piece you want to move with. It will highlight the piece and all it's possible moves.



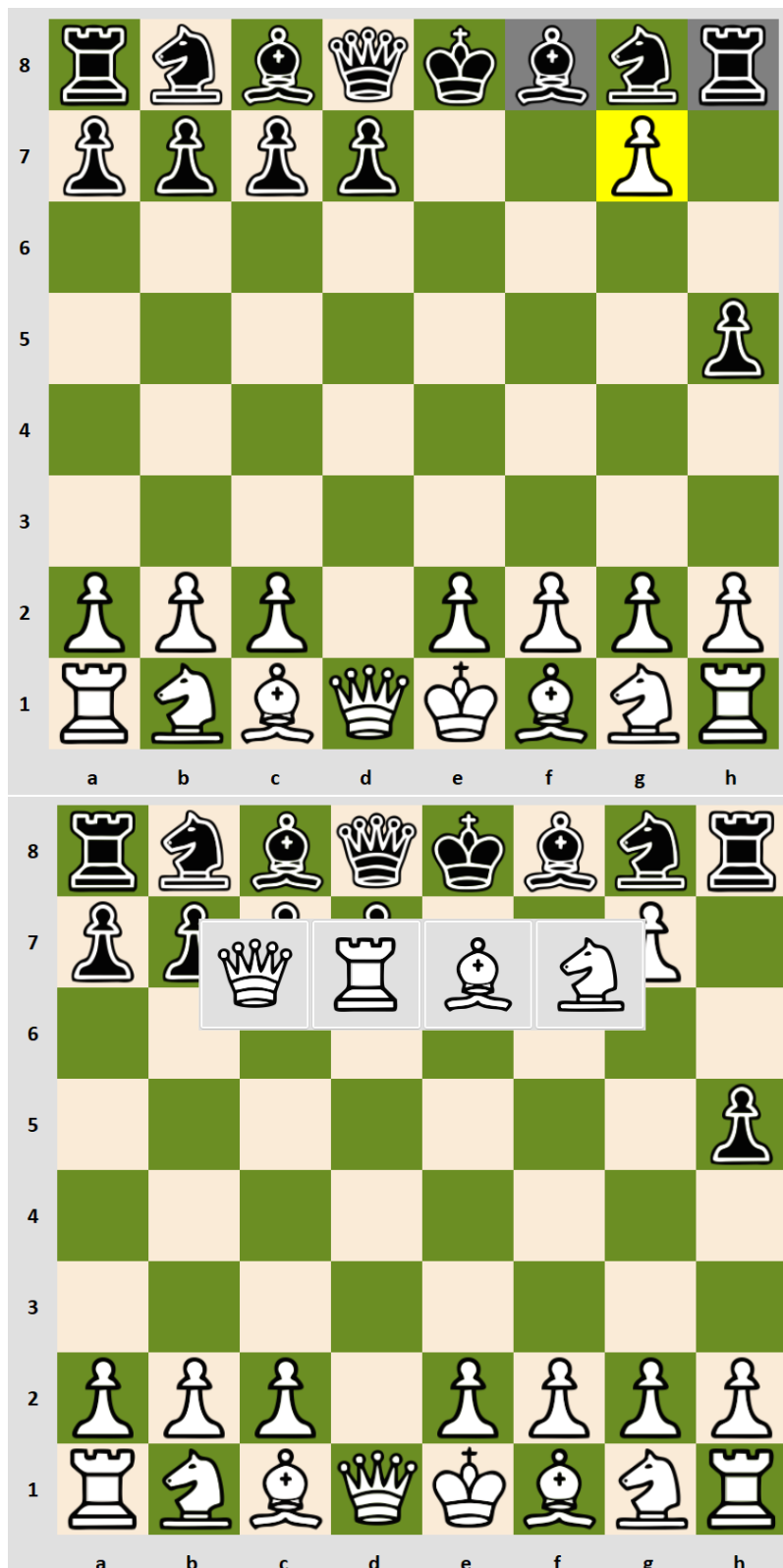
If you click on one of the gray highlighted cells, it will make a move – move the highlighted piece to the clicked cell and highlight the made move.

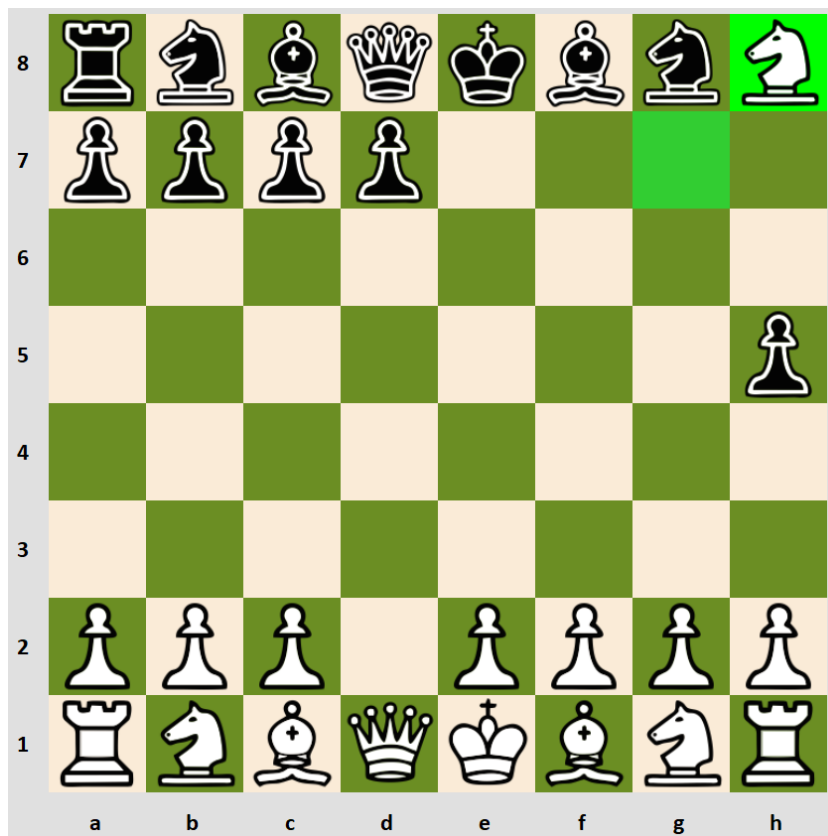


Promotion:

To promote a pawn you need to get a pawn to either the 1st or the 8th line.

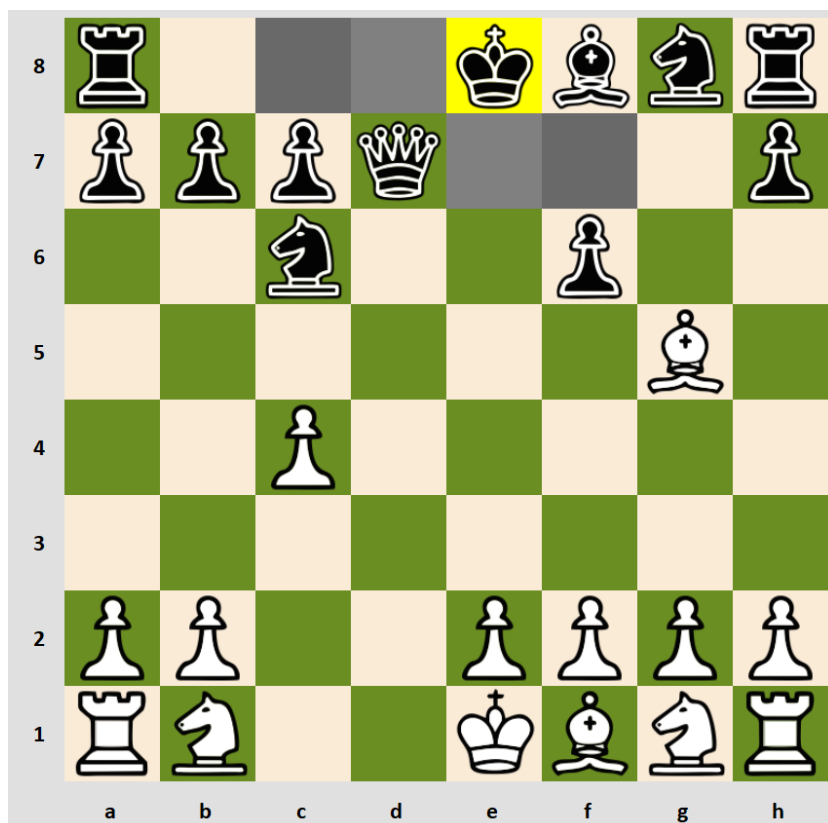
When you click on the last line a pawn can move, 4 promotion buttons will appear with the picture of pieces you can promote to. Click on one of the pieces and the will will promote to that particular piece.



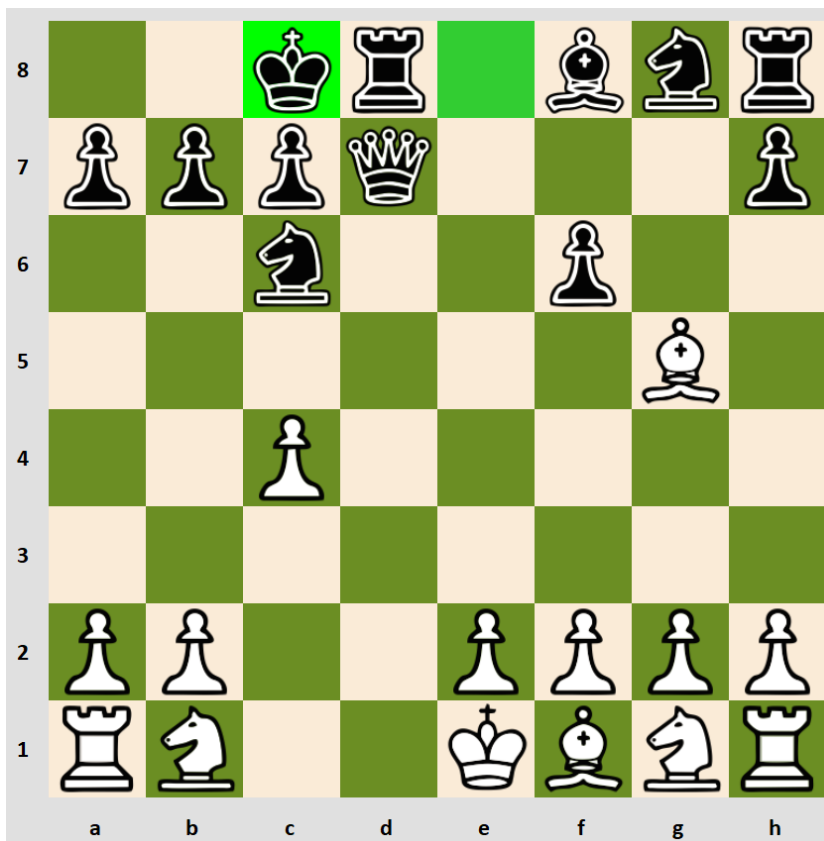


Castling:

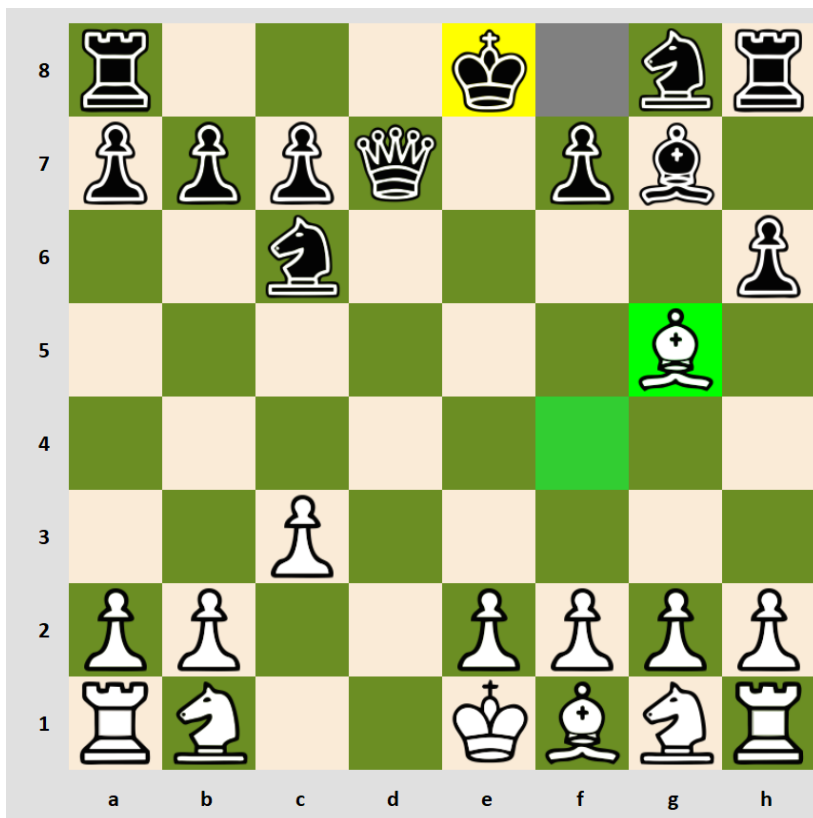
King can normally move only one piece away from his position. If all conditions to Castle are met however, one more possible move will appear that is two cells away.



If you click on that cell – it will castle the king with the rook.

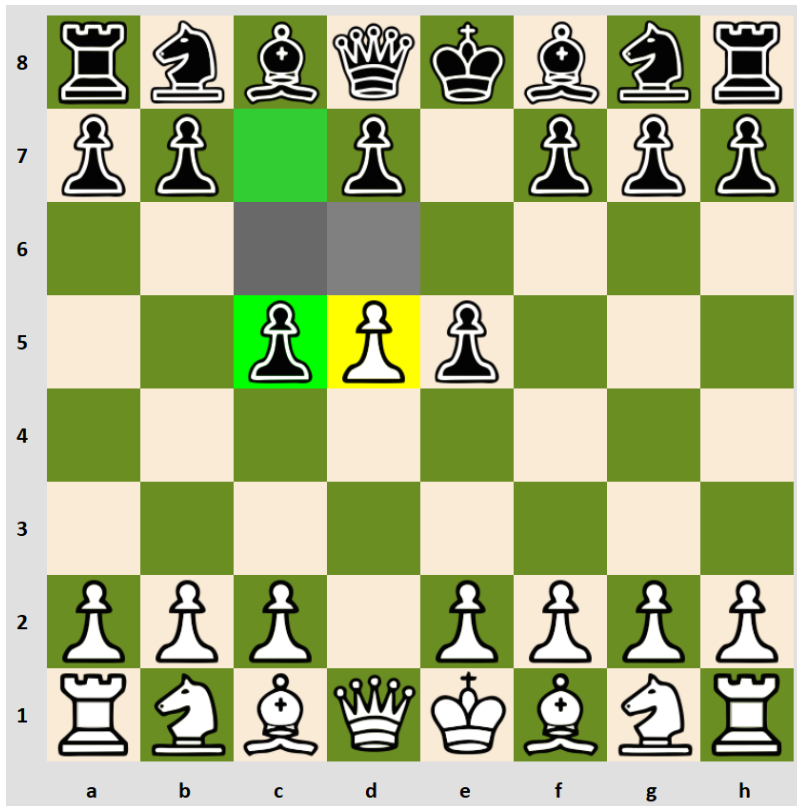


Here is an example where not all conditions are met – the cells between the rook and the king are being attacked by the opponent:

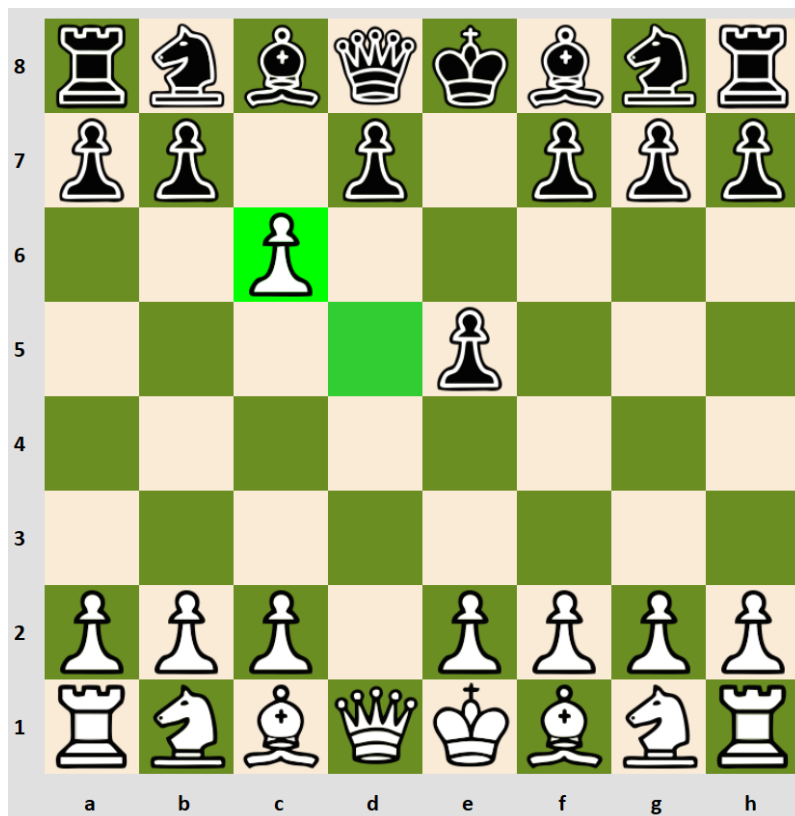


En-passant:

Similarly, if all En-passant conditions are met, a new otherwise impossible move for a pawn will appear:

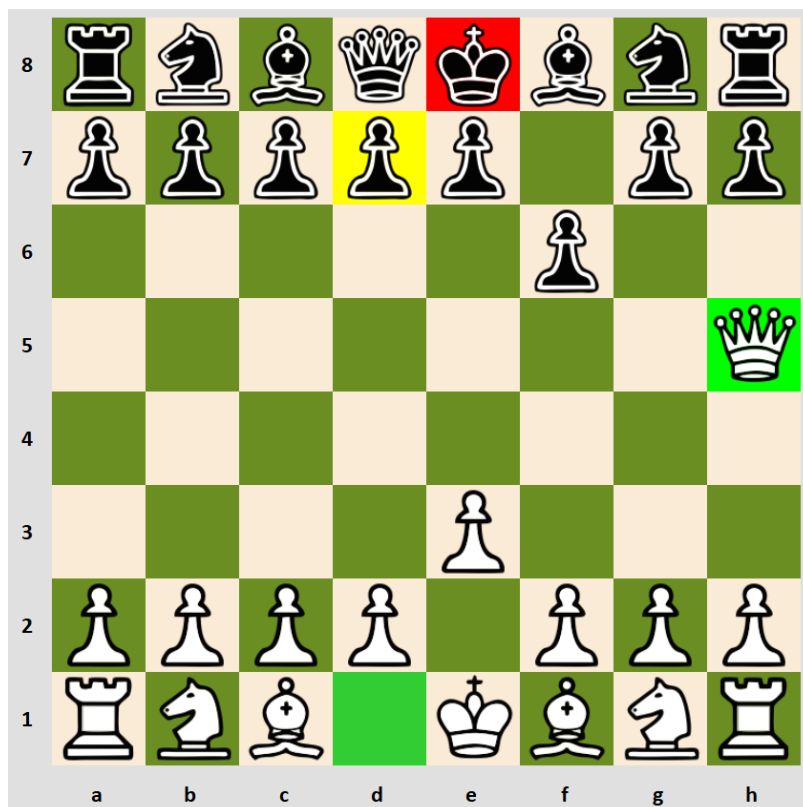


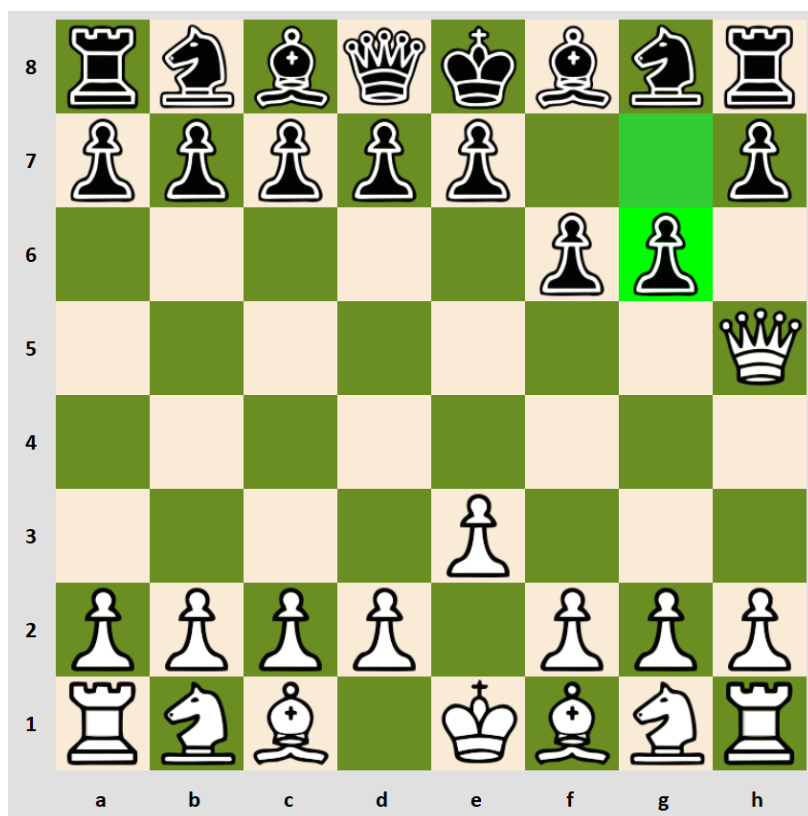
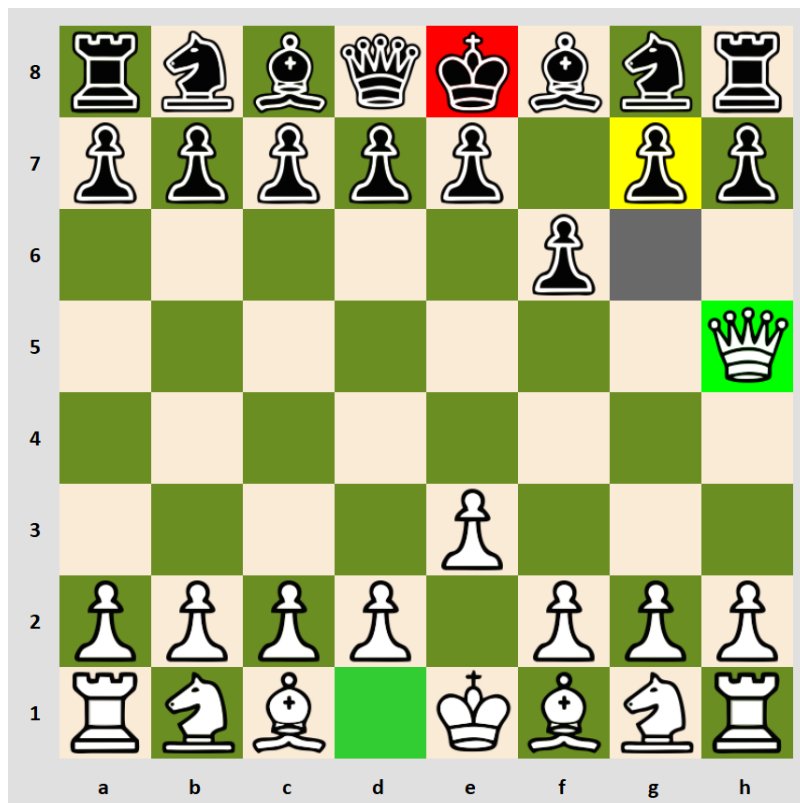
If you click on the diagonal possible move, it will make an en-passant and take the moved black pawn aswell.



Check:

If a king is in check he will be highlighted in red. If your king is in check, you can't make a move that doesn't stop the check. The only possible moves are the ones that stop the check.



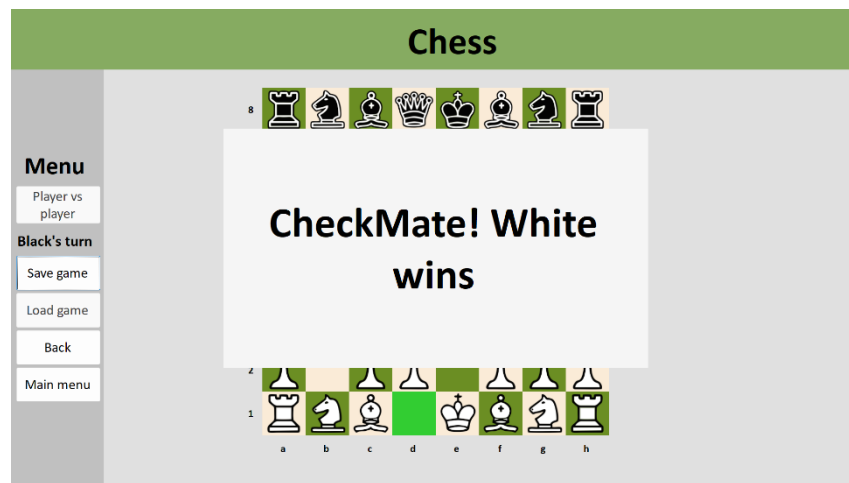


Game end:

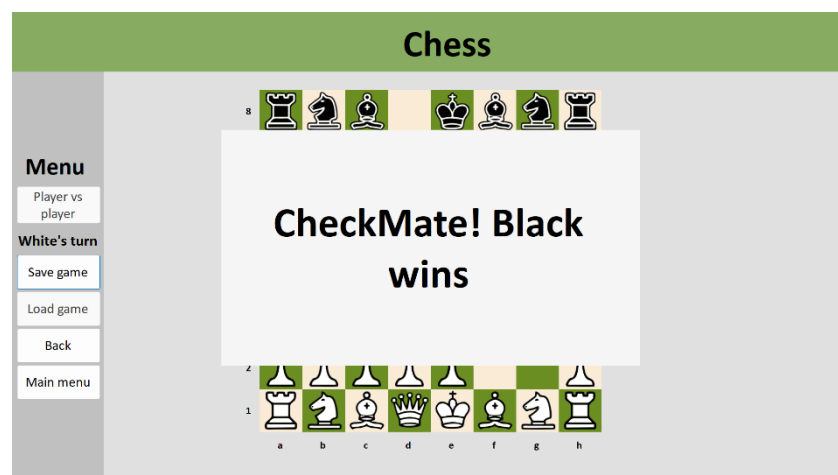
There are several ways the game can end – White wins, Black wins or a draw. When one of these 3 results happens, a big label will appear on the screen saying the outcome of the game.

White and black can win by checkmating the opponent, whereas a draw happens either when there is not enough material for either of the players to mate or if one of the players has no possible moves and it's their turn.

White wins:



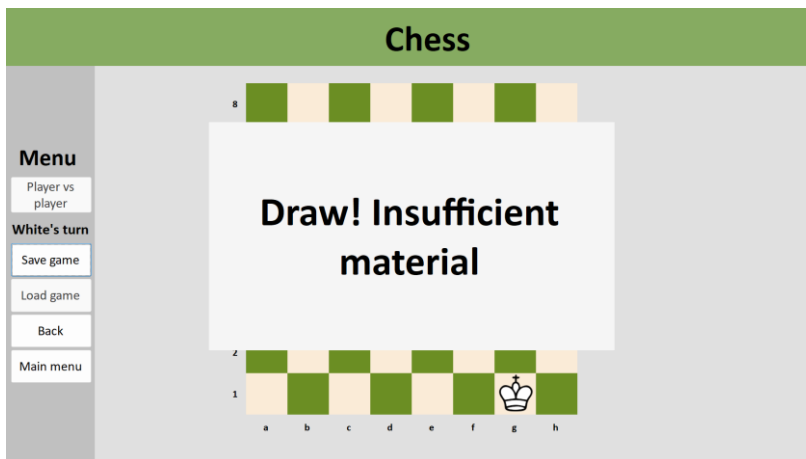
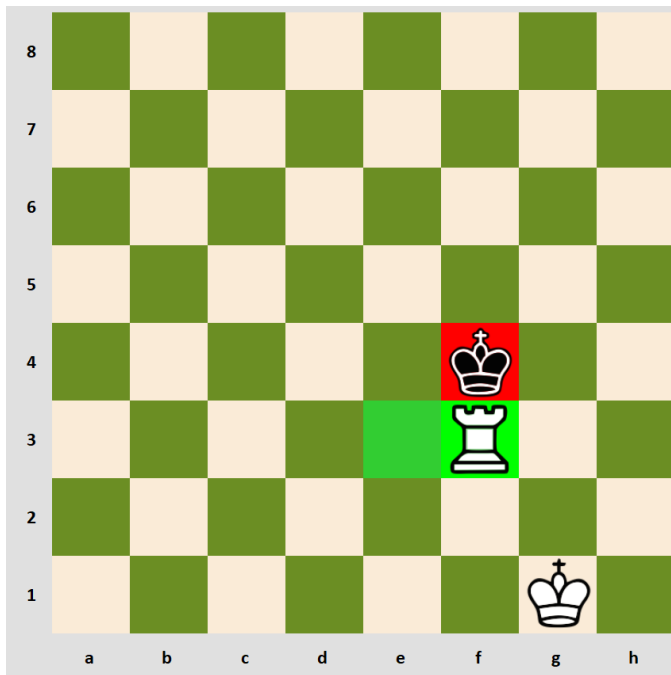
Black wins:



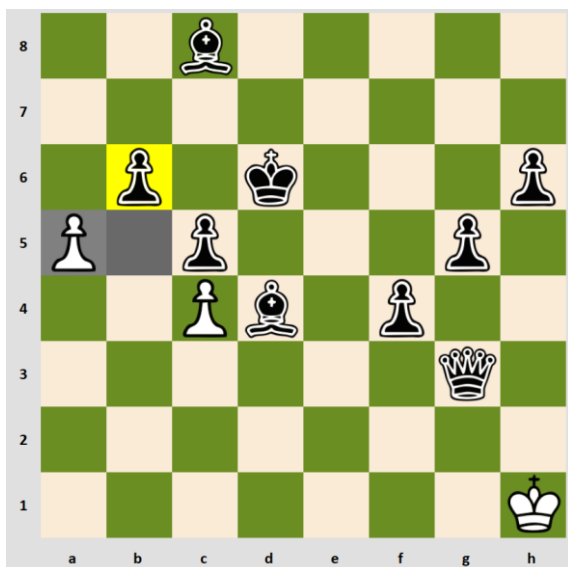
Draw:

- Insufficient material
 - o If both the players have any combination of:
 - Just the king left
 - King and a bishop
 - King and two knights

The game will result in a draw:



- Stalemate



Chess

Menu

Player vs AI

AI difficulty: 4

Save game

Load game

Back

Main menu

8



2

1

a

b

c

d

e

f

g

h

Stalemate! Draw